DS_A1_Q2

Dongju Ma

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1 Q2. (a) Fair coin

Since we are using the fair coin with tossed to the outcome of half heads and tails, we can easily get the confusion matrices with n times of experiments like Table 1, as we make an assumption that heads are true, and the observed proportion of truth is $p, (0 \leq p \leq 1)$.

1.1 (i) The expected sensitivity

So we could calculate the expected sensitivity using below formula,

$$\text{Sensitivity} = \frac{TP}{TP + FN} = \frac{\frac{np}{2}}{\frac{np}{2} + \frac{np}{2}} = \frac{1}{2}$$

1.2 (ii) The expected specificity

Also we could calculate the expected specificity by

$$\text{Specificity} = \frac{TN}{TN + FP} = \frac{\frac{n(1-p)}{2}}{\frac{n(1-p)}{2} + \frac{n(1-p)}{2}} = \frac{1}{2}$$

2 Q2. (b) Biased Coin

Still we should figure out the confusion matrices first, as the biased coin have a probability of $q, (0 \leq q \leq 1)$ to be tossed with heads outcomes. This time we still have observed proportion $p, (0 \leq q \leq 1)$ of heads, so the matrices should be like Table 2.

This time we could calculate the biased sensitivity and specificity by

$$\text{Sensitivity} = \frac{TP}{TP + FN} = \frac{nqp}{nqp + np(1-q)} = \frac{nqp}{np} = q$$
$$\text{Specificity} = \frac{TN}{TN + FP} = \frac{n(1-q)(1-p)}{nq(1-p) + n(1-q)(1-p)} = \frac{n(1-q)(1-p)}{n(1-p)} = 1 - q$$

Table 1: Confusion Matrix of Fair Coin.

	Truth Positive	Truth Negative
Predicted Positive	$\frac{np}{2}$	$\frac{n(1-p)}{2}$
Predicted Negative	$\frac{np}{2}$	$\frac{n(1-p)}{2}$

Table 2: Confusion Matrix of Biased Coin.

	Truth Positive	Truth Negative
Predicted Positive	nqp	$nq(1-p)$
Predicted Negative	$np(1-q)$	$n(1-q)(1-p)$

3 Q2. (c) Conclusions

We could make following conclusions from these results,

- If the truth condition happens with probability $P, (0 \leq P \leq 1)$, the sensitivity would be P while the specificity should be $1 - P$.

- The sum of sensitivity and specificity should be 1.